

Instructions for transcription of files with pretrained model

In this manual we will use a Google Drive account to train a speech recognition model using the [Wav2Vec2 XLSR-53](#) pretrained model as a base. At the time of writing, this is still the best model for speech recognition with a language with very little data (an **under-resourced**, or **low-resource** language).

These are the steps we will follow:

- Steps 1 – 3: Install the training code in your Google Drive
- Steps 4 – 7, 14: Export transcriptions from ELAN
- Steps 8 – 13: Process ELAN files to create training data
- Step 15: Examine transcriptions for code-switching and errors
- Steps 16 - 18: Create training and testing datasets
- Steps 19 - 24: Train the speech recognition model (30-45 mins)
- Steps 25 - 35: Use the model to transcribe a new video file.
- Steps 36 - 37: Use the model to transcribe your own voice.

The manual will ask you to download some audio and ELAN files. You can also download them all as a bundle [here](#):

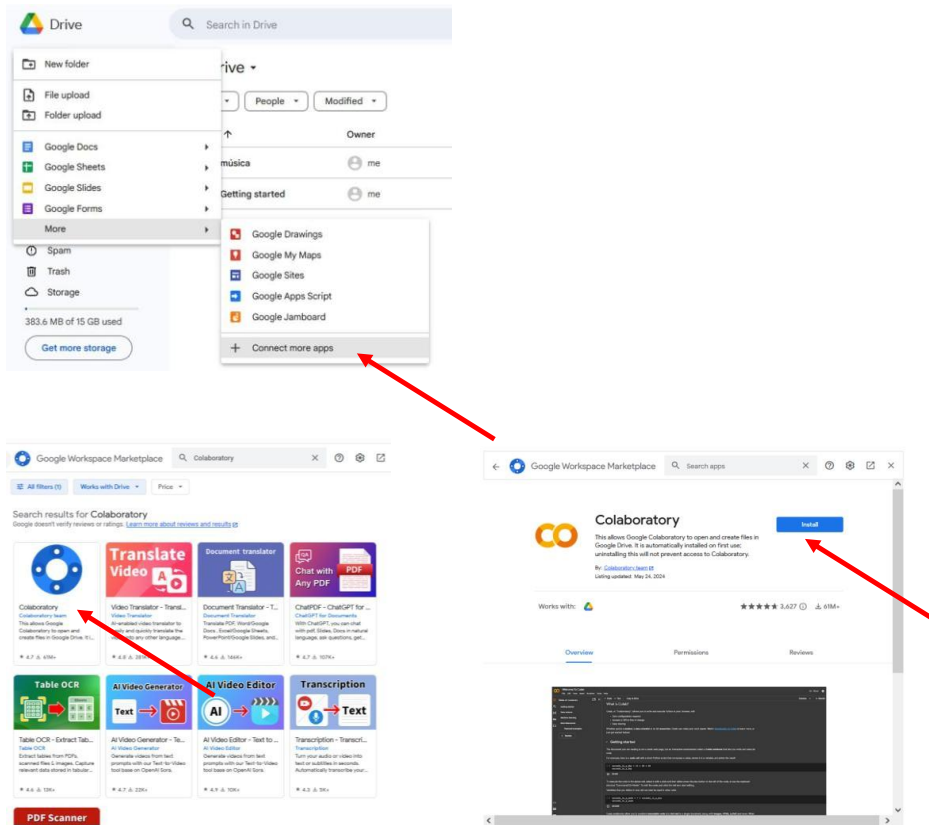
[CIM audio examples](#)

If you don't have data for your language, or would prefer not to use it for now, you can use these [Bribri audio and ELAN examples](#) for a second exercise.

(If you are in the in-person workshop, we also have USB drives with the data, in case the download is slow).

- (1) If you have **never used Google Colab**: Go to your Google Drive (with the account you will use). Go to **+New > More > Connect More Apps**. Type "colab" to find the application "Colaboratory". If it is not installed, install it.

The account should have at least 4GB of available space. If you don't have enough free space available, you can create a new account and use it for this exercise.



- (2) **If you have not used ELAN before**, you might need to install ELAN:
Download: <https://archive.mpi.nl/tla/elan/download>

 ELAN ▾ Forums Help ▾ Login MPI Archive »

Browse Archive Browse by ▾

Search 

ELAN / DOWNLOAD

Download

Intro Release notes Documentation **Download** Citing Support Resources

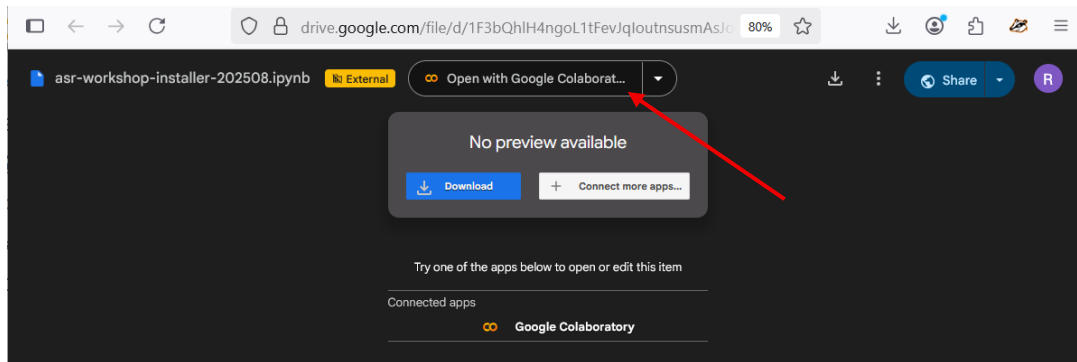
Download the current version of **ELAN** for your platform

By downloading and using ELAN you agree to cite **ELAN** in any publication based on research in which **ELAN** has been used. You can cite a paper or cite as software, e.g. as follows:
"ELAN (Version 7.0) [Computer software]. (2025). Nijmegen: Max Planck Institute for Psycholinguistics. Retrieved from <https://archive.mpi.nl/tla/elan>"
By downloading and using ELAN you also indicate you read and agree to the [terms of use and privacy policy of ELAN](#).

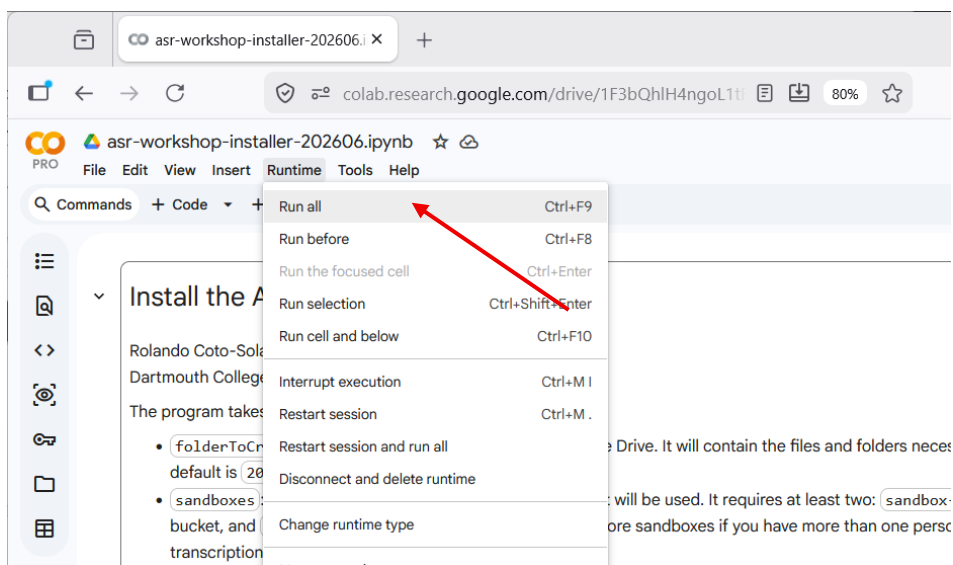
OS	Download	Format	Description
 Windows	ELAN 7.0 Simple-ELAN 1.5	[.exe] [.zip] [.msi] [.exe] [.zip]	Installer (.exe [a Setup wizard with options] and .msi [silent installation for all users]) and portable, unpack-and-run (.zip) versions all include a Java 24 runtime. Requires: 64-bit, Windows 11, 10, (8 not tested anymore)
 macOS	ELAN 7.0 ELAN 7.0-M1 Simple-ELAN 1.5 Simple-ELAN 1.5-M1	[.zip] [.dmg] [.zip] [.dmg] [.zip] [.dmg] [.zip] [.dmg]	.zip and .dmg files containing a .app which includes a Java 24 runtime (Java 21 for Intel based Macs). The M1 version only works on <i>Apple silicon</i> Macs (including M2 etc.). Requires: 64-bit, macOS High Sierra (10.13) or higher for Intel Macs, macOS Big Sur (11) or higher for Mx Macs.
 Linux	ELAN 7.0 Simple-ELAN 1.5	[.tar.gz] [.deb] [.tar.gz] [.deb]	Archive files (.tar.gz) or Debian installers (.deb), all including a Java 24 runtime. Requires: 64-bit



- (3) Go to this address: [ASR Installer](#) (open it in a new tab). This should open a file with code (a Jupyter Notebook) in Google Colab. The file should be called asr-workshop-installer-202606.ipynb. If what you see is a bunch of gibberish code, or a message that says that there is “No Preview Available”, then click on *Open with* and open this file with Google Colab.



Once it's open, go to *Runtime > Run all*. The system might ask for your permission to write files onto your Google Drive. Please grant those permissions. (See step 11 below for more details). This will install the files of the speech recognition workflow.¹



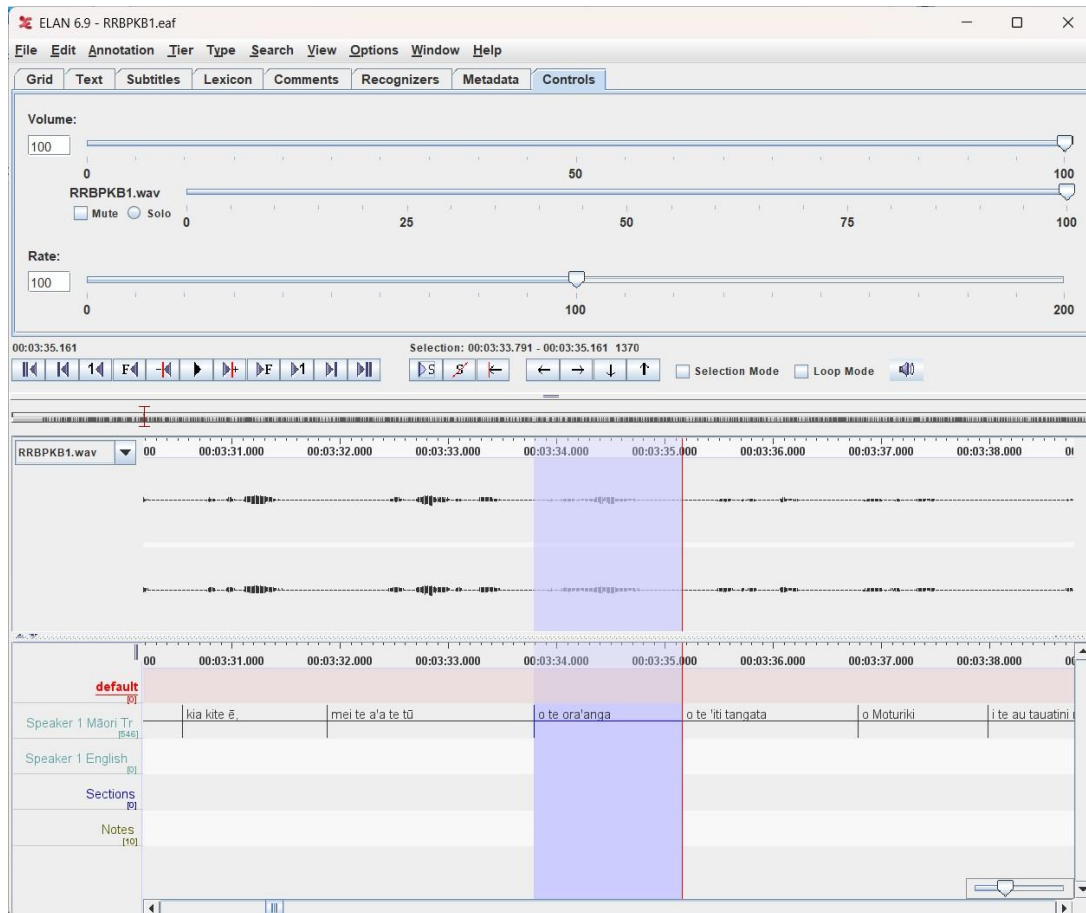
¹ Just in case, take a look at the last cell. If it says that there are errors locating the files, just run it again. (Sometimes the creation of the Google Docs spreadsheets takes a bit of time).

- (4) First, let's prepare some files for the training. Download these files:

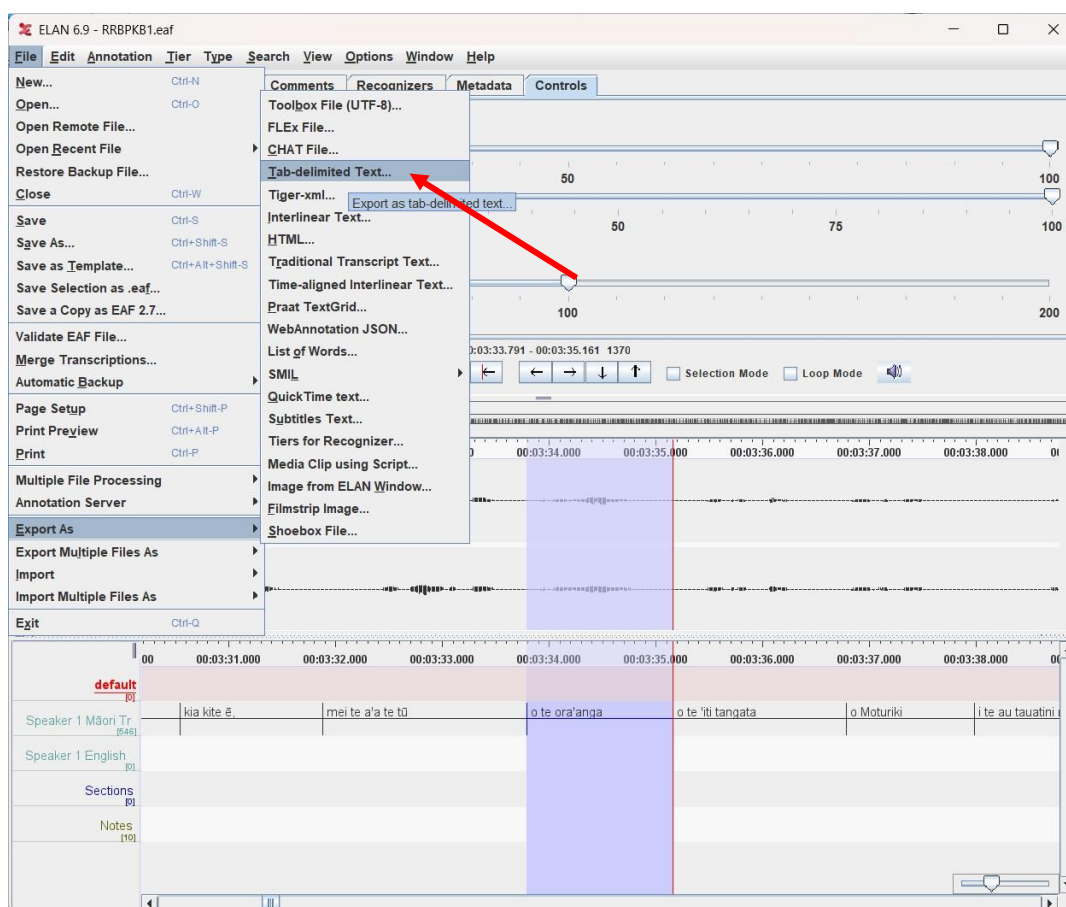
ELAN: [Zip with file RRBPKB1.eaf](#)

Wave: [RRBPKB1.wav](#)

- (5) Open the EAF file using ELAN. You should be able to see a window like the one below.



(6) Please go to *File > Export As > Tab-delimited Text...*



- (7) We'll export the transcription tier. Go to *File > Export As > Tab-delimited Text*. Select only one of the tiers with the transcriptions. (You need to export each speaker separately). At the bottom of the form, select *Begin time*, *End time*, *Duration* and *ss.msec*. (**Don't select hh:mm:ss.ms**). Click on "OK" and save it as `RRBPKB1-bpt.txt`. (Those are the initials of this speaker).

Export as tab-delimited text

Export tier(s) as tab-delimited text

Select tiers

By Tier Names By Types By Participants By Annotators By Languages

☐ default
☒ Speaker 1 Māori Transcription
☐ Speaker 1 English Translation
☐ Sections
☐ Notes

☐ Show only root tiers

⬆ ⬇ A-Z Undo Sort Select All Select None

Output options

☐ Restrict to selected time interval
☐ Add master media time offset to annotation times
☐ Include header lines containing media file information
☐ Exclude tier names from output
☐ Exclude participant names from output
☐ Separate column for each tier
☒ Repeat values of annotations spanning other annotations
☐ Only repeat within annotation hierarchies
☐ Sliced annotation output showing temporal co-occurrences
☐ Include the annotation id
☐ Include description from the controlled vocabulary

Include time column for:

☒ Begin Time
☒ End Time
☒ Duration

Include time format:

☐ hh:mm:ss.ms
☒ ss.msec
☐ msec
☐ SMPTE Timecode (hh:mm:ss)

OK Close

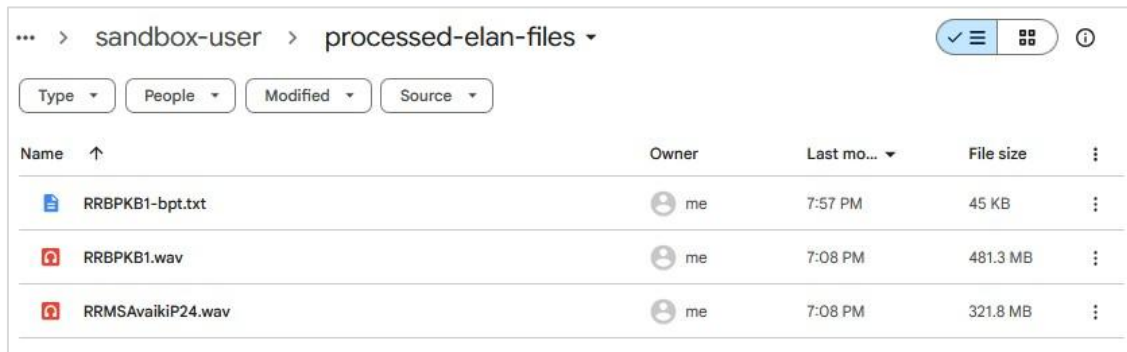
The saved file should look like this. It has five columns: (i) The name of the tier, (ii) the name of the speaker, (iii) the start time of each segment, (iv) the end time of each segment, and (v) the transcription of the words in each time segment.

1	Speaker 1 Māori Transcription	BPT	48.123	51.396	Te 'eu'eu'anga i tō tātou 'ai tūpuna
2	Speaker 1 Māori Transcription	BPT	51.396	53.634	Nā Damon Salesa
3	Speaker 1 Māori Transcription	BPT	53.634	56.753	'uri'anga 'e Terangi Nikora
4	Speaker 1 Māori Transcription	BPT	56.753	60.306	I te mata'iti 2002,
5	Speaker 1 Māori Transcription	BPT	60.306	66.686	kua kitea'ia teta'i anga ūpoko takere i Viti.
6	Speaker 1 Māori Transcription	BPT	66.686	71.015	Nō teta'i va'ine waitata ra rāi ē kua pou
7	Speaker 1 Māori Transcription	BPT	71.015	77.015	te toru tauatini mata'iti te roa mei tōna mate'anga
8	Speaker 1 Māori Transcription	BPT	77.015	79.464	Ko 'ai rā tēia va'ine?
9	Speaker 1 Māori Transcription	BPT	79.464	83.247	Mei te a'a ra te tū o tōna ora'anga?

- (8) Please go to your Google Drive and upload the text file onto the folder
202606-cim-asr/sandbox-user/processed-elan-files

You also need to upload your audio file here (in this case, RRBPKB1.wav). I have preuploaded it so you don't have to use time to do this at the workshop, but please remember to upload the audio file you will be using.

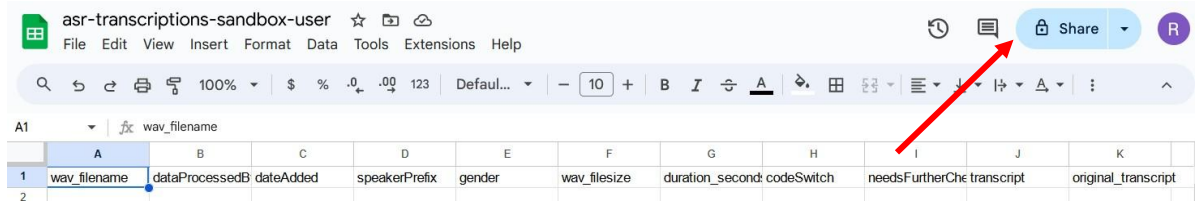
The processed-elan-files folder should now look like this:



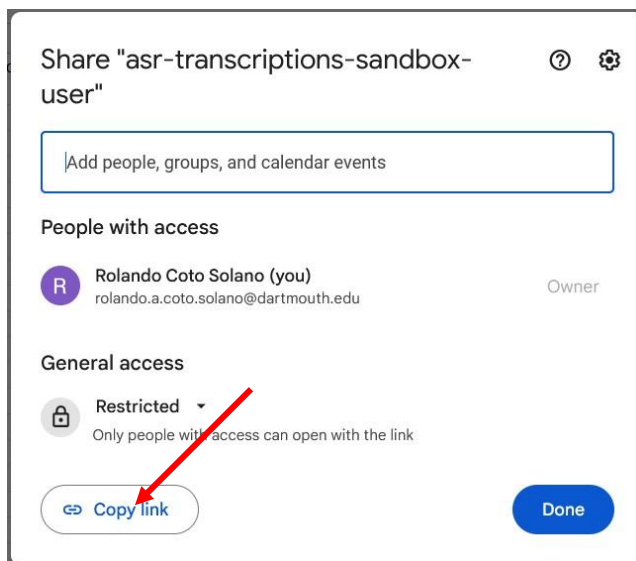
The screenshot shows a Google Drive interface for the folder 'processed-elan-files' within the 'sandbox-user' directory. The interface includes a breadcrumb trail, filter buttons for 'Type', 'People', 'Modified', and 'Source', and a table of files. The table has columns for 'Name', 'Owner', 'Last modified', and 'File size'. Three files are listed: 'RRBPKB1-bpt.txt' (45 KB), 'RRBPKB1.wav' (481.3 MB), and 'RRMSAvaikiP24.wav' (321.8 MB), all owned by 'me'.

Name	Owner	Last modified	File size
RRBPKB1-bpt.txt	me	7:57 PM	45 KB
RRBPKB1.wav	me	7:08 PM	481.3 MB
RRMSAvaikiP24.wav	me	7:08 PM	321.8 MB

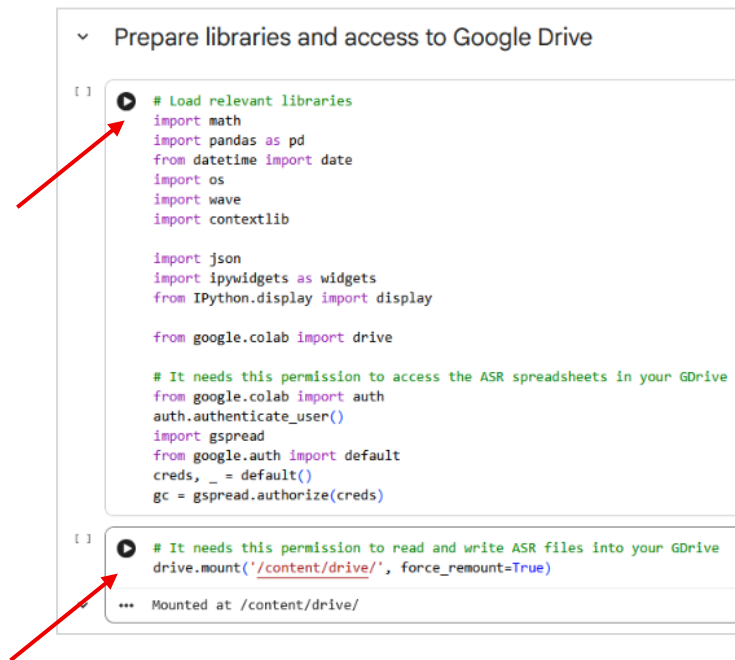
- (9) Go to the folder that was created during the installation (202606-cim-asr). You will find a Google Sheet called `asr-transcriptions-sandbox-user`. Open it. You will see a file with names of columns.



Click on the “Share” button on the upper right. Click on “Copy link”. This will copy the address of the Google sheet, which we will need later. (I suggest you put this in a text file so you don’t lose the URL). Then, click on “Done”.



- (10) Go to the notebook `from-elan-to-wav-and-gsheet.ipynb`. Run the first two boxes of code (the ones with the title “Prepare libraries and access to Google Drive”) by moving your cursor to the box and clicking on the “Play” button.



```
[ ] ▶ # Load relevant libraries
import math
import pandas as pd
from datetime import date
import os
import wave
import contextlib

import json
import ipywidgets as widgets
from IPython.display import display

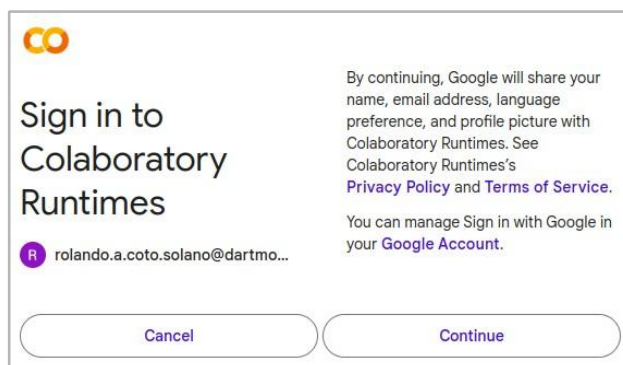
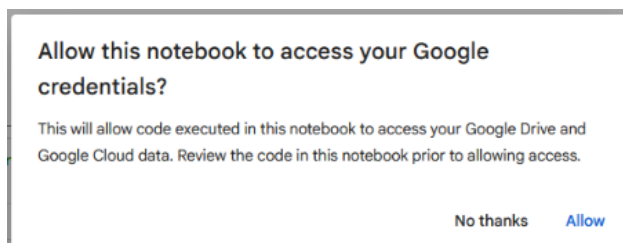
from google.colab import drive

# It needs this permission to access the ASR spreadsheets in your GDrive
from google.colab import auth
auth.authenticate_user()
import gspread
from google.auth import default
creds, _ = default()
gc = gspread.authorize(creds)

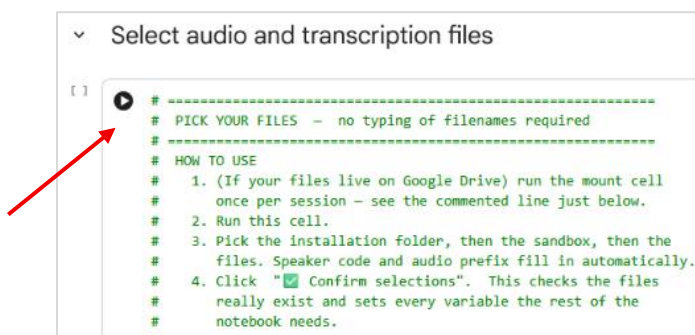
[ ] ▶ # It needs this permission to read and write ASR files into your GDrive
drive.mount('/content/drive/', force_remount=True)

*** Mounted at /content/drive/
```

This will connect the notebook to your Google Drive and have it ready to find the files it needs to process. Click on “Allow”, then select your Google account and then click on “Continue”. (It might ask you to “Continue” up to four times).

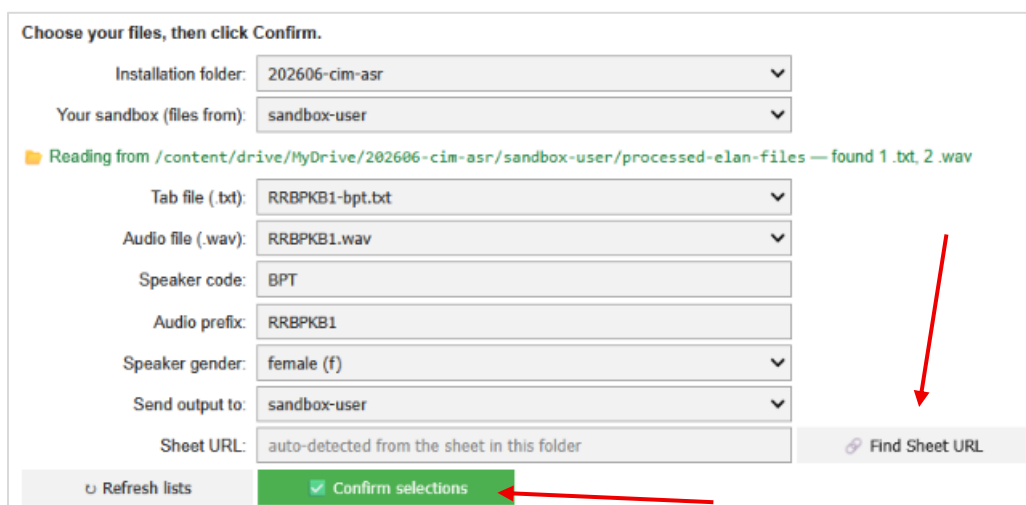


- (11) Go to the third box of code (“Select audio and transcription files”) and click on the play button to run the code. Then, scroll downwards.



```
# =====  
# PICK YOUR FILES - no typing of filenames required  
# =====  
# HOW TO USE  
# 1. (If your files live on Google Drive) run the mount cell  
# once per session - see the commented line just below.  
# 2. Run this cell.  
# 3. Pick the installation folder, then the sandbox, then the  
# files. Speaker code and audio prefix fill in automatically.  
# 4. Click "Confirm selections". This checks the files  
# really exist and sets every variable the rest of the  
# notebook needs.  
# =====
```

This will start a series of boxes that you can use to select the files you want to process. Make sure you are on the folder where you installed the code (e.g. 202606-cim-asr), select your transcription file (e.g. RRBPKB1-bpt.txt) and your audio file (e.g. RRBPKB1.wav). Change the gender to “male”. Then, click on “Find sheet URL”. This will get the address of the Google Sheet whose access you authorized on step 9. Once this is done, click on “Confirm selection”.



Choose your files, then click Confirm.

Installation folder: 202606-cim-asr

Your sandbox (files from): sandbox-user

Reading from /content/drive/MyDrive/202606-cim-asr/sandbox-user/processed-elan-files — found 1 .txt, 2 .wav

Tab file (.txt): RRBPKB1-bpt.txt

Audio file (.wav): RRBPKB1.wav

Speaker code: BPT

Audio prefix: RRBPKB1

Speaker gender: female (f)

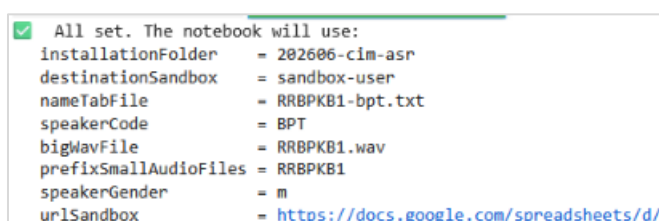
Send output to: sandbox-user

Sheet URL: auto-detected from the sheet in this folder

Find Sheet URL

Refresh lists Confirm selections

When you click on “Confirm selection”, you should get an output like the one below. The `urlSandbox` should be a full address that matches the address you copied on step 9.



```
✓ All set. The notebook will use:  
installationFolder = 202606-cim-asr  
destinationSandbox = sandbox-user  
nameTabFile = RRBPKB1-bpt.txt  
speakerCode = BPT  
bigWavFile = RRBPKB1.wav  
prefixSmallAudioFiles = RRBPKB1  
speakerGender = m  
urlSandbox = https://docs.google.com/spreadsheets/d/
```

- (12) Run the rest of the blocks. The file should have 546 valid annotations. By the end of the notebook you should see something like this:

```
Total rows before new data: 1
Total rows after new data: 547
Added rows: 546
```

- (13) If you go to the folder 202508-cim-asr, and then go to the spreadsheet asr-transcriptions-sandbox-user, you will see the transcriptions and their information (e.g. wav file, duration, speaker, etc).

	A	B	C	D	E	F	G	H	I	J	K
	wav_filename	dataProcessedB	dateAdded	speakerPrefix	gender	wav_filesize	duration_second	codeSwitch	needsFurtherChe	transcript	original_transcript
1	BPT-RRBPKB1- user		20240620	BPT	m	104652	3.2679375			te 'eue'anga i tē Te 'eue'anga i tō ta	
2	BPT-RRBPKB1- user		20240620	BPT	m	71460	2.2306875			nā damon salesē Nā Damon Salesa	
3	BPT-RRBPKB1- user		20240620	BPT	m	100104	3.1258125			'un'anga 'e terang 'un'anga 'e Terangi	
4	BPT-RRBPKB1- user		20240620	BPT	m	113746	3.552125			i te mata'iti 2002 I te mata'iti 2002,	
5	BPT-RRBPKB1- user		20240620	BPT	m	204222	6.3795			kua kitea'ia teta'i kua kitea'ia teta'i an	
6	BPT-RRBPKB1- user		20240620	BPT	m	138752	4.3335625			nō teta'i va'ine va Nō teta'i va'ine vaiat	
7	BPT-RRBPKB1- user		20240620	BPT	m	191948	5.9959375			te toru tauatini m te toru tauatini mat	
8	BPT-RRBPKB1- user		20240620	BPT	m	78280	2.4438125			ko 'ai rā tēia va'in Ko 'ai rā tēia va'ine?	
9	BPT-RRBPKB1- user		20240620	BPT	m	121474	3.793625			mei te a'a ra te ti Mei te a'a ra te tū o	
10	BPT-RRBPKB1- user		20240620	BPT	m	68278	2.13125			ē	E,
11	BPT-RRBPKB1- user		20240620	BPT	m	127840	3.9925625			i 'akape'ea ra tōn i 'akape'ea ra tōna 'e	
12	BPT-RRBPKB1- user		20240620	BPT	m	47362	1.477625			i te 'aere mai'ang i te 'aere mai'anga	
13	BPT-RRBPKB1- user		20240620	BPT	m	67824	2.1170625			mei atia	mei Atia
14	BPT-RRBPKB1- user		20240620	BPT	m	46000	1.4350625			i te pae tonga	i te Pae Tonga

- (14) Repeat steps 5 through 8 with the EAF file whose name starts with RRMSAvaikiP24:

ELAN: [Zip with file RRMSAvaikiP24.eaf](#)

Wave: [RRMSAvaikiP24.wav](#)

Export the annotations from the tier “Speaker 1 Transcription” and save them into the file RRMSAvaikiP24-msc.txt. Repeat steps 10 to 13 to process the data. These should be the inputs you use:

```
installationFolder: 202606-cim-asr
destinationSandbox: sandbox-user
nameTabFile: RRMSAvaikiP24-msc.txt
speakerCode: MSC
bigWavFile: RRMSAvaikiP24.wav
prefixSmallAudioFiles: RRMSAvaikiP24
speakerGender: f
```

By the end of this procedure, you should have about 727 rows in the sandbox spreadsheet. Their total duration is approximately 46 minutes. (You can check this by adding up the numbers on column G of the spreadsheet: It should add up to 2773 seconds).

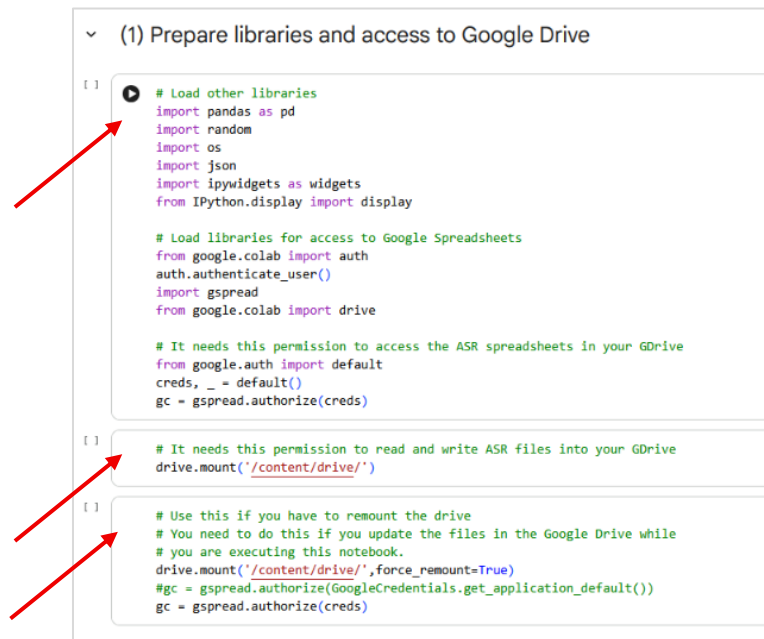
```
Total rows before new data: 547
Total rows after new data: 727
Added rows: 180
```

- (15) Go to the Google Drive (folder: 202508-cim-asr) and then to the spreadsheet with the data (asr-transcriptions-sandbox-user) and check the transcriptions. Are there any of them with code-switching, or any other issues? If so, please tag them. (For example, row 548 has English in it. It needs to be tagged as codeSwitch; please add a 1 to that cell: H548).

asr-transcriptions-sandbox-user

	A	B	C	D	E	F	G	H	I	J	K	L
1	wav_filename	dataProcessedB	dateAdded	speakerPrefix	gender	wav_filesize	duration_second	codeSwitch	needsFurtherChc	transcript	original_transcript	
548	MSC-RRMSAvaiki user		20240620	MSC	f	319254	9.97425		1	eng excuse me	<ENG>Excuse me	
549	MSC-RRMSAvaiki user		20240620	MSC	f	156938	4.901875			te ika a mau	TE: IKA A MAUI	
550	MSC-RRMSAvaiki user		20240620	MSC	f	289700	9.0506875			i te 'iti tangata pō	I te 'iti tangata Porinetia e toto'a ar	
551	MSC-RRMSAvaiki user		20240620	MSC	f	198768	6.2090625			tao mai ki te ad i	Tao mai ki te AD 800, kua kitea e	

- (16) Go to the notebook `from-gsheet-to-wav2vec2-files.ipynb`. Run the first three blocks of code (under the heading “(1) Prepare libraries and access to Google Drive”). This will connect your notebook to your Google Drive. Please authorize the access.



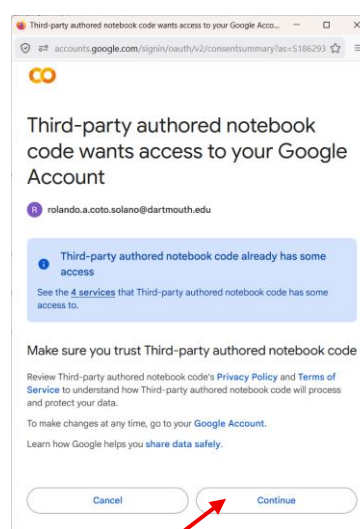
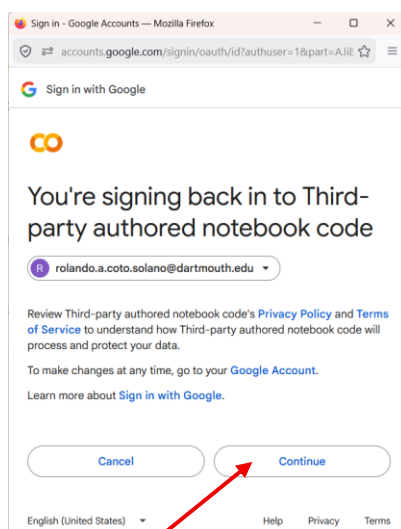
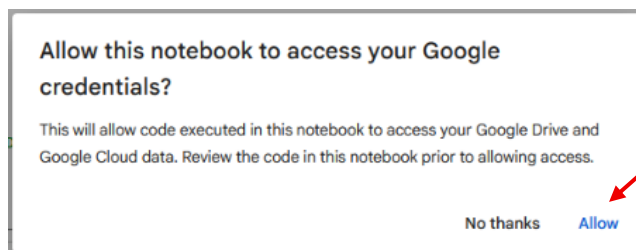
```
[ ]: # Load other libraries
import pandas as pd
import random
import os
import json
import ipywidgets as widgets
from IPython.display import display

# Load libraries for access to Google Spreadsheets
from google.colab import auth
auth.authenticate_user()
import gspread
from google.colab import drive

# It needs this permission to access the ASR spreadsheets in your GDrive
from google.auth import default
creds, _ = default()
gc = gspread.authorize(creds)

[ ]: # It needs this permission to read and write ASR files into your GDrive
drive.mount('/content/drive/')

[ ]: # Use this if you have to remount the drive
# You need to do this if you update the files in the Google Drive while
# you are executing this notebook.
drive.mount('/content/drive/', force_remount=True)
#gc = gspread.authorize(GoogleCredentials.get_application_default())
gc = gspread.authorize(creds)
```



(17) Run the fourth box of code. Press the black “play” button to run it.



```
(2) Fill out metadata

# -----
# SPECIFY FILES TO MAKE TRAINING FILES
# -----
# Reads the same Google Sheet as the previous notebook (detected
# automatically), then you choose the partition / model options.
# Click "Confirm selections" to set every variable the rest of
# this notebook needs (and to check the split adds up to 100%).
#
# You normally won't edit anything here: the dropdowns list the
# folders in your MyDrive. If your Drive lives somewhere unusual,
# adjust MYDRIVE_ROOT below.
# -----

# ----- Config -----
MYDRIVE_ROOT = "/content/drive/MyDrive" # where installation folders live
DEFAULT_INSTALLATION_FOLDER = "202606-cim-asr" # default selection if present
SHEET_NAME_PREFIX = "asr-transcriptions-" # sheet = <prefix><sandbox>
SOFTWARE_OPTIONS = ["wav2vec2", "ds", "whisper", "wavlm"]
```

Scroll down to find the dialog box with the options to build the training files. Please click on “Find Sheet URL” to get the Google Sheet with your data. (This is the URL from step 9). You probably don’t have to modify anything else. When you’re done, click on “Confirm selections”.



1 · Data source

Installation folder: 202606-cim-asr

Sandbox (destinationSandbox): sandbox-user

Sheet URL (urlSandbox): auto-detected from the sheet in this folder

Find Sheet URL

pathAudioFilesInTraining = /content/drive/MyDrive/202606-cim-asr/sandbox-user/wav/

2 · Model & data options

Software: wav2vec2

File prefix (filePrefix): cim-wav2vec2

Max WAV duration (s): 15

☐ Use code-switched data

☐ Use doubtful data

3 · Train / validation / test split

Train %: 80

Validation %: 10

Test %: 10

split adds up to 100% ✓

Confirm selections

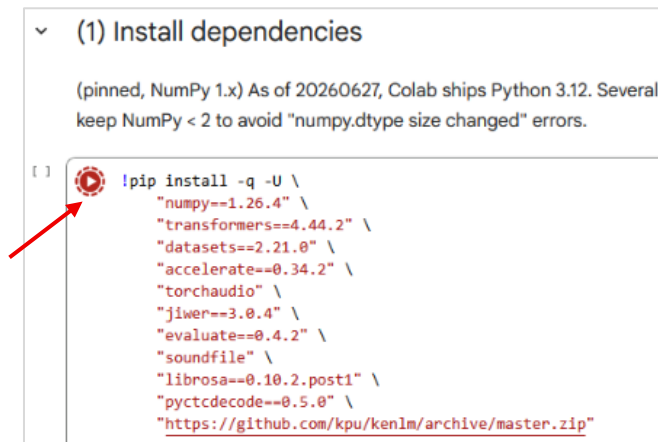
The output should look like this:

```
✓ All set. The notebook will use:
installationFolder = 202606-cim-asr
destinationSandbox = sandbox-user
urlSandbox = https://docs.google.com/spreadsheets/d/16mLHkoiwBj_uB7ISw
useCodeSwitchedData = 0
useDoubtfulData = 0
percentageTrainSet = 80
percentageValidSet = 10
percentageTestSet = 10
maxwavDuration = 15
software = wav2vec2
filePrefix = cim-wav2vec2
pathAudioFilesInTraining = /content/drive/MyDrive/202606-cim-asr/sandbox-user/wav/
```

- (18) Run all of the code blocks in the notebook. It will split the data into three randomly-assigned sets: training (80%), validation (10%) and testing (10%). At the end there will be three new files in your sandbox:

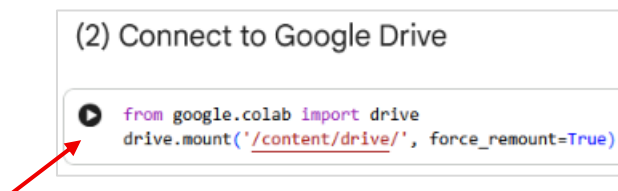
```
cim-wav2vec2-train.csv  
cim-wav2vec2-valid.csv  
cim-wav2vec2-test.csv
```

- (19) Go to the notebook `trainer-w2v2.ipynb`. Run the first box (“Install dependencies”). Don’t worry if it says that “the session crashed”. This is both expected and necessary.²



```
(1) Install dependencies  
  
(pinned, NumPy 1.x) As of 20260627, Colab ships Python 3.12. Several  
keep NumPy < 2 to avoid "numpy.dtype size changed" errors.  
  
[ ] !pip install -q -U \  
    "numpy==1.26.4" \  
    "transformers==4.44.2" \  
    "datasets==2.21.0" \  
    "accelerate==0.34.2" \  
    "torchaudio" \  
    "jiwer==3.0.4" \  
    "evaluate==0.4.2" \  
    "soundfile" \  
    "librosa==0.10.2.post1" \  
    "pyctcdecode==0.5.0" \  
    "https://github.com/kpu/kenlm/archive/master.zip"
```

- (20) Run the second code box to connect to Google Drive. Please authorize the connection.



```
(2) Connect to Google Drive  
  
[ ] from google.colab import drive  
    drive.mount('/content/drive/', force_remount=True)
```

² We need this to properly load all the packages that the trainer needs to function. In particular, the version of NumPy is different from the one provided by Google Colab at the time of writing.

(21) Run the third box of code to select the files that you will train on.

(3) Select the files to train on

```
# =====  
# SELECT TRAINING FILES AND SET UP TRAINING  
# =====  
# Run this cell, choose the installation folder and sandbox, and  
# the train / validation / test CSVs found in that sandbox are  
# offered in dropdowns (auto-matched by name, override if needed).  
# Set the language prefix, run id and training options, then click  
# "Confirm selections".  
#  
# You normally won't edit anything here. If your Drive lives  
# somewhere unusual, adjust MYDRIVE_ROOT below.  
# =====
```

You probably don't need to modify anything here. Make sure you have a `train` file for the training, a `valid` (or `dev`) for the validation, and a `test` for testing.

1 - Which files to train on

Installation folder: 202606-cim-asr

Sandbox (currentSandbox): sandbox-user

Found 3 CSV(s) in /content/drive/MyDrive/202606-cim-asr/sandbox-user

Train CSV (trainFile): cim-wav2vec2-train.csv

Validation CSV (validFile): cim-wav2vec2-valid.csv

Test CSV (testFile): cim-wav2vec2-test.csv

2 - Training options

Language prefix (asrLang): cim

Run ID (runId): 01

Train epochs: 34

KenLM n-grams: 4

Confirm selections

When you are ready, click on "Confirm selections". The output should look like this:

```
✓ All set. The notebook will use:  
installationFolder = 202606-cim-asr  
currentSandbox     = sandbox-user  
trainFile          = cim-wav2vec2-train.csv  
validFile          = cim-wav2vec2-valid.csv  
testFile           = cim-wav2vec2-test.csv  
asrLang            = cim  
runId              = 01  
desiredTrainEpochs = 34  
ngrams             = 4
```

- (22) Run cells 4 to 13. The example will take about 30~45 minutes, but the training with the full dataset could take several hours, depending on how much data you have and how many epochs you specified.

The program will tell you how many steps it has left, and long it estimates the training will take. As the training progresses you will see messages about the **Training Loss** and the **Evaluation Loss**. This is the accumulated error for the training and validation set. In this example, the loss will be calculated every 100 steps³.

You will also see numbers for the **CER** (character error rate) and the **WER** (word error rate) of the validation/dev set. This is the percentage of characters and words that are spelled wrong in the automatic transcription, compared to the gold standard. Doing this is like a test while we are training, so that we can diagnose how the training is doing.

[1224/1224 27:27, Epoch 33/34]

Step	Training Loss	Validation Loss	Wer	Cer
100	0.451500	0.482752	0.429293	0.131437
200	0.437100	0.473597	0.377104	0.127836
300	0.379300	0.473931	0.350168	0.111991
400	0.306000	0.584145	0.383838	0.124235
500	0.270700	0.493017	0.345118	0.110911
600	0.230600	0.533456	0.335017	0.108030
700	0.205800	0.492785	0.329966	0.105149
800	0.176000	0.530048	0.309764	0.102989
900	0.160600	0.567885	0.316498	0.108750
1000	0.146500	0.549253	0.301347	0.102269
1100	0.127100	0.580505	0.301347	0.100828
1200	0.120500	0.553337	0.296296	0.096507

Keep an eye on the **Evaluation Loss**. This number should decrease as the training continues. If it starts creeping up, you might be **overfitting** the model. This means that the model is starting to memorize the patterns in the training set and failing to generalize for the recordings in the validation set. (Look at how, after step 600, the validation loss starts to increase. This tells you that, at this point, the model is starting to overfit).

³ What is the relationship between epochs and steps?

$$\text{epochs} = (\text{desiredSteps} * \text{batchSize} * \text{gradientAccumulationSteps}) / \text{trainingSamples}$$

In this example, batchSize = 8, gradientAccumulation = 2, trainingSamples = 580 (the number of audio files in the training set). As for the desired steps, our 4-hour Cook Islands Maori set trains in about 4000 steps. The 16-hour Marquesan dataset trains in about 10000 steps.

- (23) Run cells 14 to 17. This will run the evaluation of the checkpoints. It tells you the **CER** (character error rate) and the **WER** (word error rate). As the training progresses, the error decreases. The one you should focus on the the **median CER (KenLM)** and the **median WER (KenLM)**.

```
=====
'final' model promoted from: checkpoint-1224 (median WER+KenLM = 0.1667
=====

ASR language: cim
Condition: wav2vec2 | Run: 01 | Epochs: 34
Base model: facebook/wav2vec2-large-xlsr-53
KenLM: 4-gram (lm-cim-4-correct.arpa)
Checkpoints evaluated: 2
Test sentences per checkpoint: 73

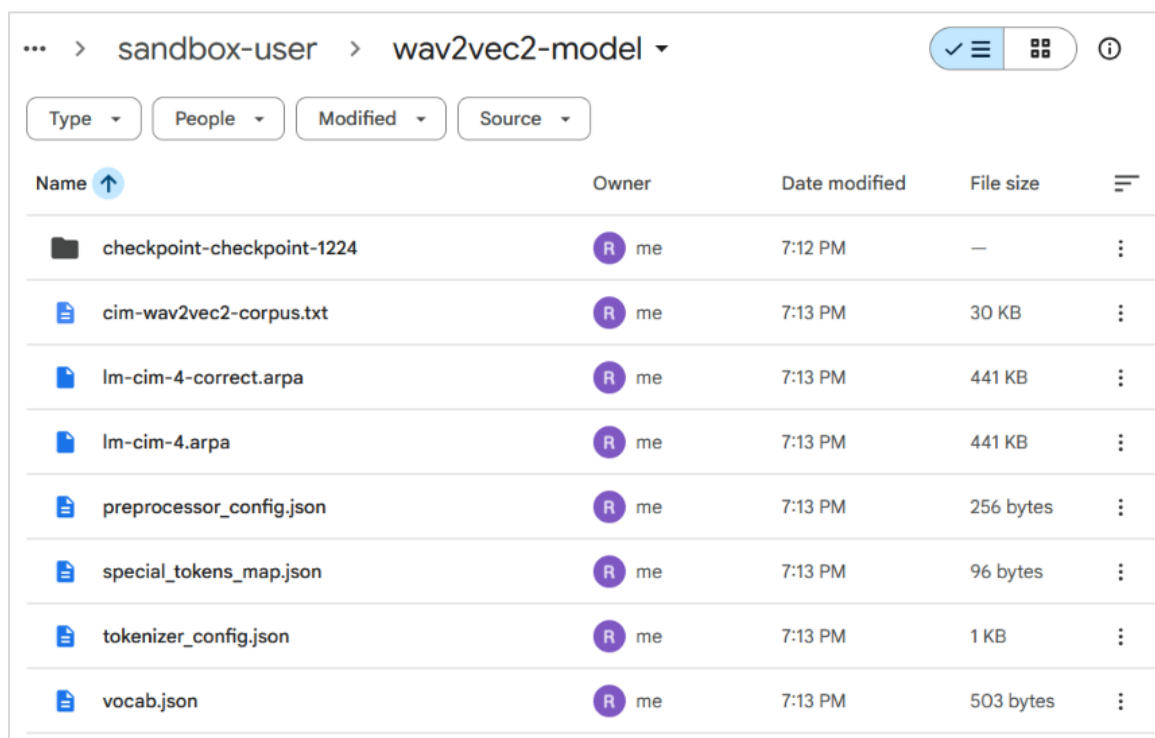
===== FINAL / BEST MODEL =====
(The 'final' saved model has been set to the checkpoint with the
 best KenLM performance.)
'final' is a copy of: checkpoint-1224
Median WER (no LM) : 0.2500
Median CER (no LM) : 0.0588
Median WER (KenLM) : 0.1667
Median CER (KenLM) : 0.0494
Agg WER (KenLM) : 0.2010
Agg CER (KenLM) : 0.0737

===== ALL CHECKPOINTS =====
(sorted best -> worst by KenLM performance)

--- [rank 1] checkpoint-1224 <-- promoted to 'final' ---
MEDIAN (per-sentence):
  WER no-LM: 0.2500 CER no-LM: 0.0588
  WER KenLM: 0.1667 CER KenLM: 0.0494
AGGREGATE (corpus-level):
  WER no-LM: 0.2855 CER no-LM: 0.0930
  WER KenLM: 0.2010 CER KenLM: 0.0737
FIRST TEST SENTENCE:
  GOLD : ko tā chris 'anga'anga tēia
  NO-LM: ko tāchris 'anga'anga tēia
  KenLM: ko tā chris 'anga'anga tēia
```

- (24) Run cell 18. This will save the best model onto your Google Drive and leave it there for use in the future steps. We chose to save only one model, the final one that was trained (checkpoint-1224). We only saved one because each of these is approximately 3.5GBs.

You can look at the files in the model if you go to the folder `wav2vec2-model`.



The screenshot shows a Google Drive interface for the folder 'wav2vec2-model' under the user 'sandbox-user'. The interface includes filters for Type, People, Modified, and Source. Below the filters is a table listing the files and folders in the directory.

Name	Owner	Date modified	File size
checkpoint-checkpoint-1224	me	7:12 PM	—
cim-wav2vec2-corpus.txt	me	7:13 PM	30 KB
lm-cim-4-correct.arpa	me	7:13 PM	441 KB
lm-cim-4.arpa	me	7:13 PM	441 KB
preprocessor_config.json	me	7:13 PM	256 bytes
special_tokens_map.json	me	7:13 PM	96 bytes
tokenizer_config.json	me	7:13 PM	1 KB
vocab.json	me	7:13 PM	503 bytes

- (25) Now, let's try to transcribe a completely new file ⁴. Take the file `kia-oranarehearsal.wav` and upload it into the Google Drive folder `202606-cim-asr/sandbox-user/audiocfiles-to-transcribe`. You will need the file on your computer for the following steps. You can download it here: <https://rcweb.dartmouth.edu/RCoto/kia-orana-rehearsal.wav>

⁴ It should be in the WAV or MP4 format, or you need to convert it to WAV or MP4. This because ELAN doesn't accept other formats.

- (26) Go to the notebook `inference-transcribe-w2v2-longRecording.ipynb`. Run the first block of code. If the computer says you have “too many sessions open”, click on “Manage sessions” and terminate the GPU sessions. Then, click on “Close” and try to run the first block of code again.

Too many sessions

You have too many active sessions. Terminate an existing session to continue.

[Cancel](#) [Manage sessions](#)

Active sessions

Runtime disconnected

Title			Last execution	
 <code>trainer-w2v2.ipynb</code>		GPU	0 minutes ago	
 <code>from-gsheet-to-wav2vec2-files.ipynb</code>			2 hours ago	

Run the second block of code to connect to the Google Drive.

- (27) Run the third block of code. You can use this to select the audio file you want to transcribe, and the checkpoint that you want to use for the transcription. You probably don't need to edit anything in this box. When you are ready, click on "Confirm selections".

1 · What to transcribe

Installation folder: 202606-cim-asr

Sandbox (currentSandbox): sandbox-user

Audio file (audioFileName): kia-orana-rehearsal.mp4

1 file(s) in /content/drive/MyDrive/202606-cim-asr/sandbox-user/audiofiles-to-transcribe

Checkpoint (modelCheckpointToUse): checkpoint-checkpoint-1224

1 checkpoint(s) in /content/drive/MyDrive/202606-cim-asr/sandbox-user/wav2vec2-model — latest selected

2 · Processing options

Min segment duration (ms): 100

Max WAV duration (s): 15

☒ Use GPU (uncheck for slower CPU)

Confirm selections

You should get output like this:

```
✓ All set. The notebook will use:
installationFolder = 202606-cim-asr
currentSandbox    = sandbox-user
audioFileName     = kia-orana-rehearsal.mp4
modelCheckpointToUse = checkpoint-checkpoint-1224
minDurationOfFile = 100
maxWavDuration    = 15
useGPU            = yes
```

- (28) Run the rest of the code. This will take a few minutes⁵. At the end of the process, the program will give you output like this:

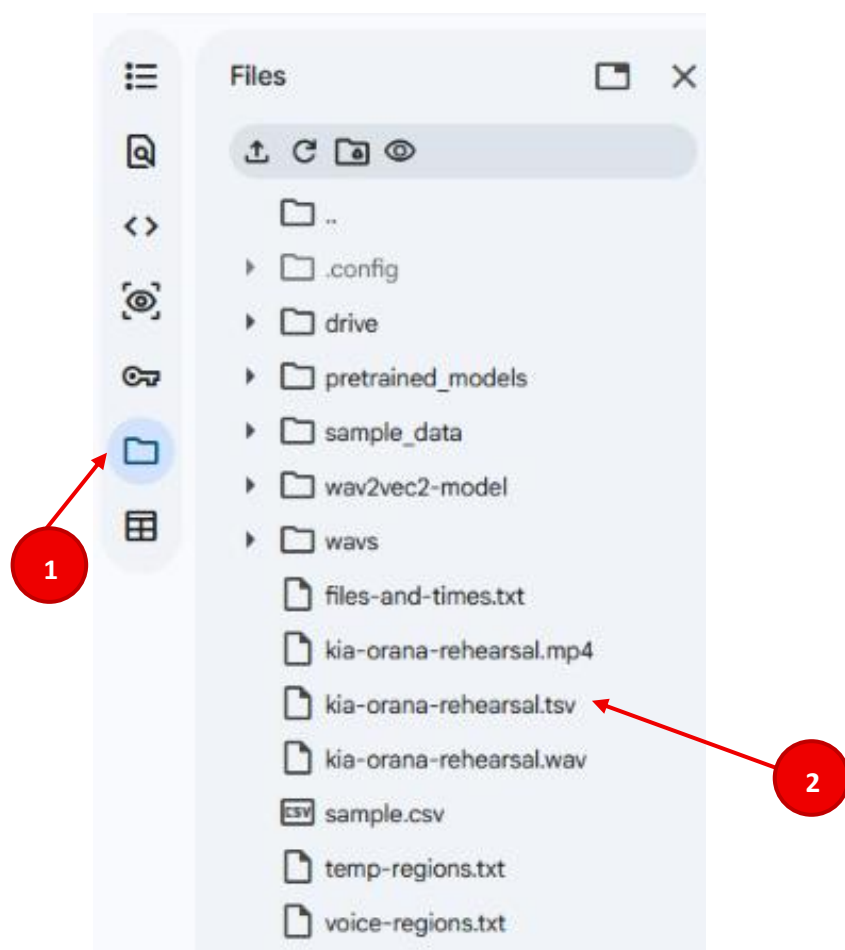
```
ki orana ki au rana apē'ēkoe
meitaki mai taki

The results are stored in:
kia-orana-rehearsal.tsv
```

This contains the transcription corresponding to the audio, and the name of a TSV file where the transcriptions are saved along with their time indeces.

⁵ The program will find the regions of the audio file that contain human speech using a package called SpeechBrain, and then split the audio file into smaller pieces so that they fit in the memory of the GPU.

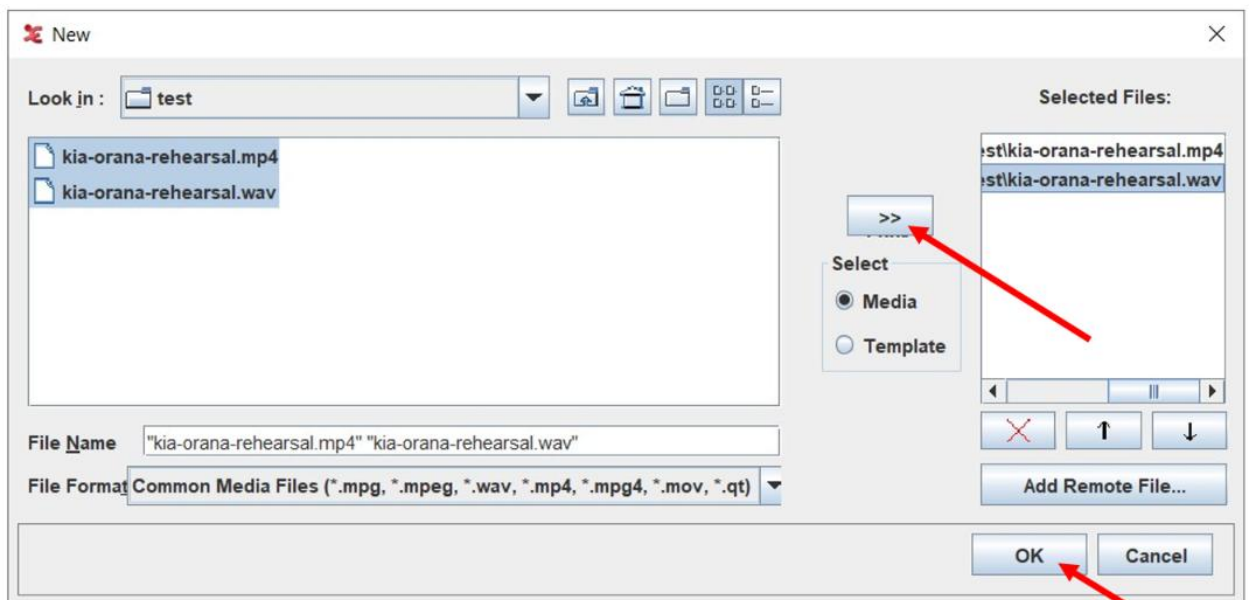
- (29) Download that TSV file. We will use it to create a new ELAN file. (You can find this by clicking the folder icon on the left of the Google Colab window. Then, right-click on the file `kia-orana-rehearsal.tsv` and download it).



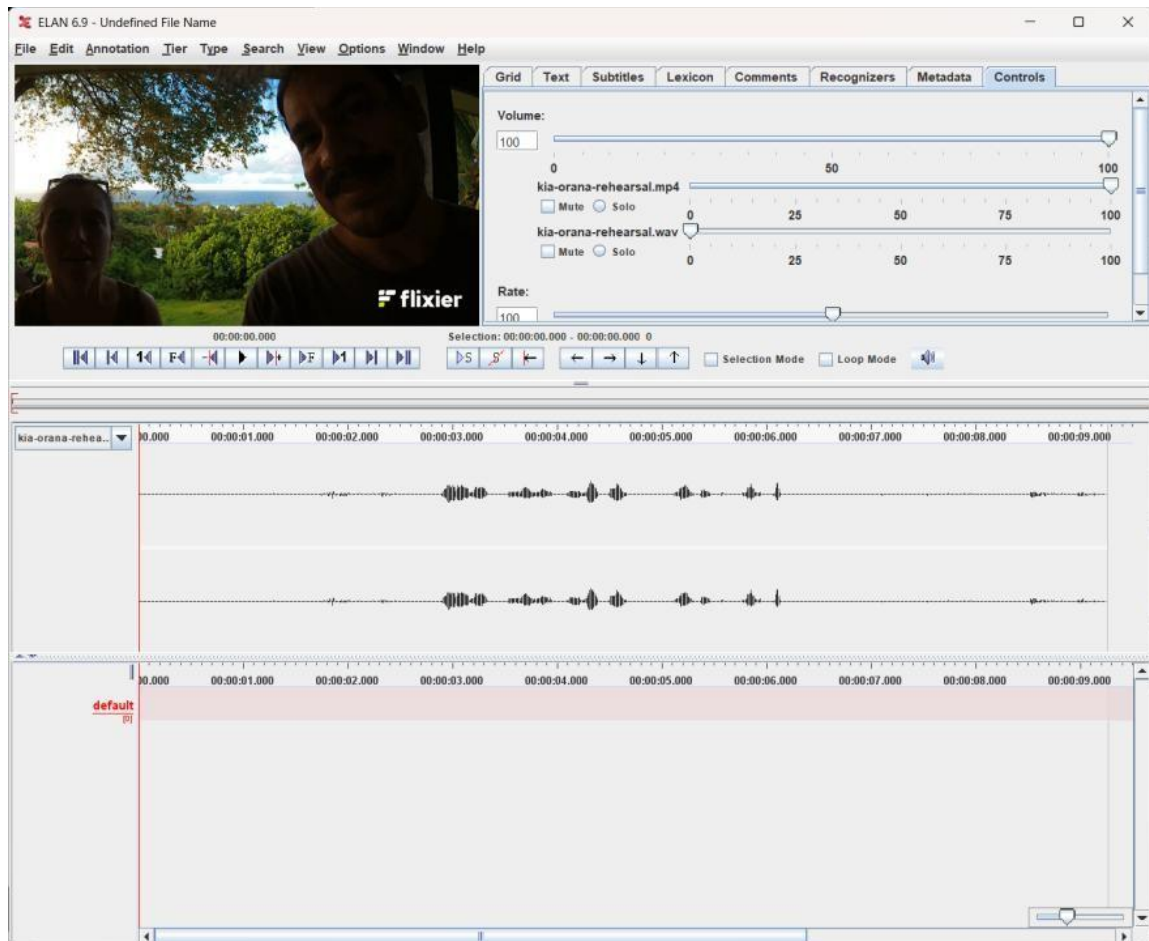
- (30) Download the following file. It's the video of the "kia orana" rehearsal.
<https://rcweb.dartmouth.edu/RCoto/kia-orana-rehearsal.mp4>
(Right click on the video and click on *Save video as* to save it onto your computer).



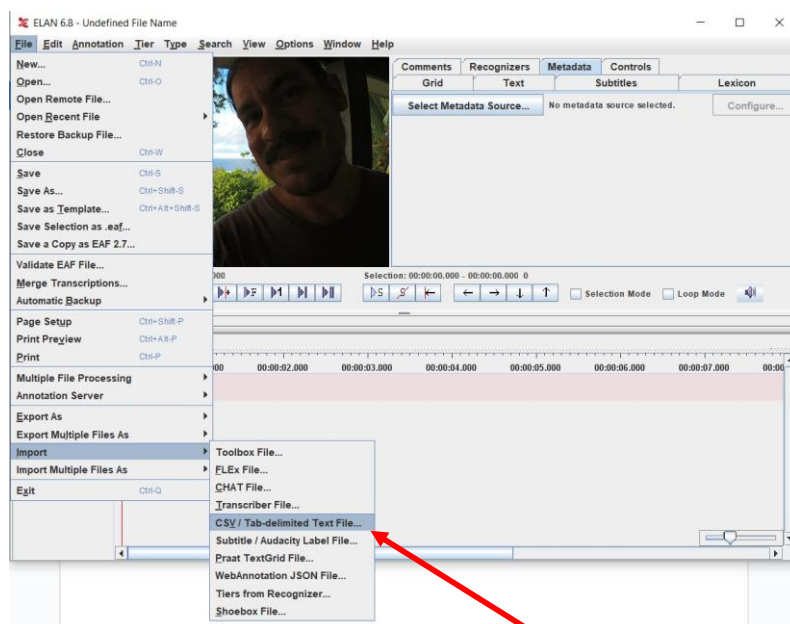
- (31) Go to ELAN and click on File > New. Look for the video file (kia-orana-rehearsal.mp4) and the audio file (kia-orana-rehearsal.wav) and click on the icon with two arrows to select the file. Then click on "OK".



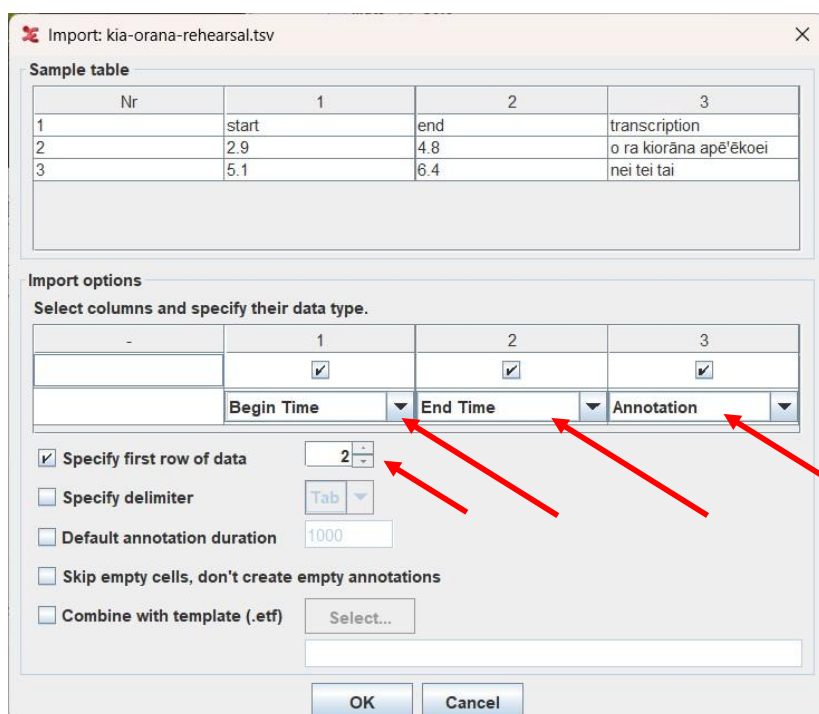
- (32) You will now see the ELAN screen with the video and the spectrogram and amplitude for the audio. **SAVE YOUR FILE.** Save it frequently to make sure you don't lose your work. ELAN is unfortunately very prone to crashing.



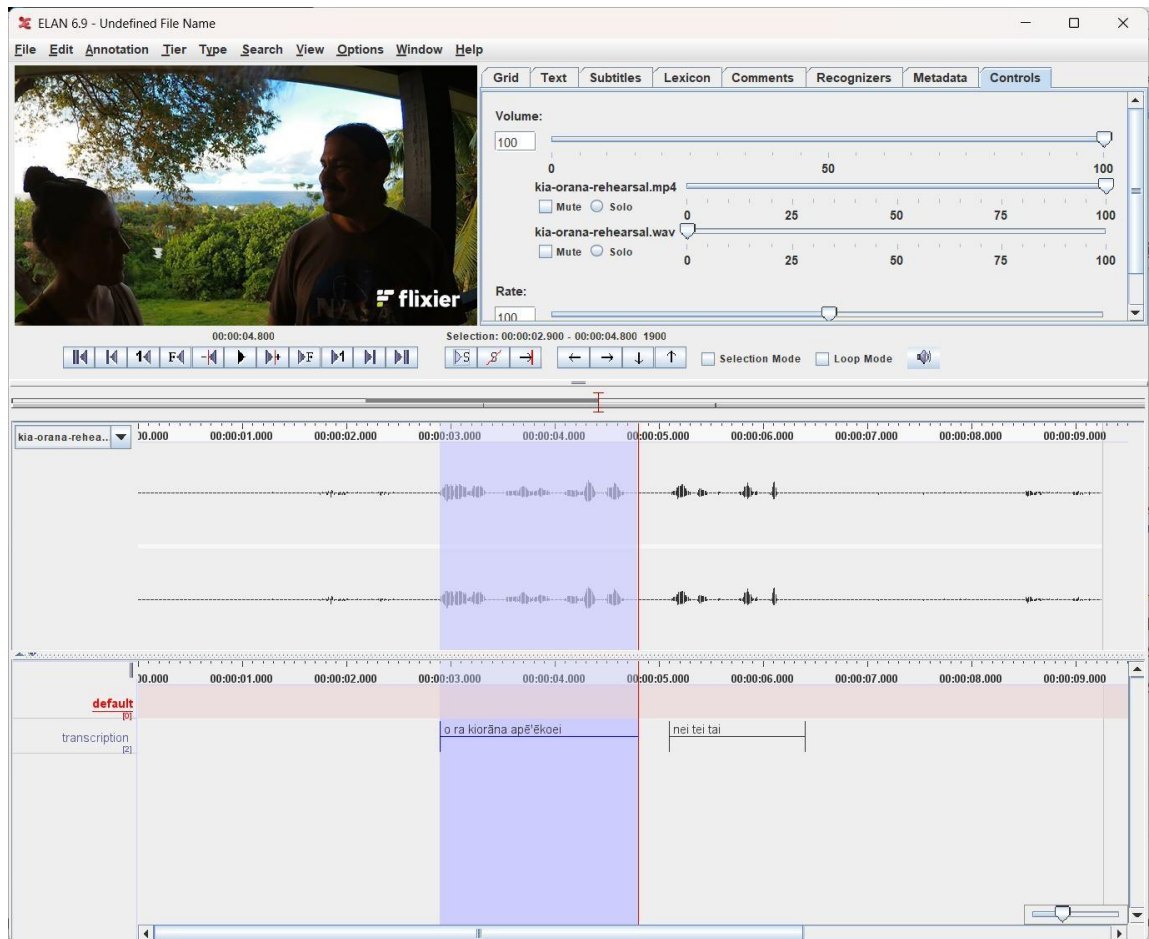
- (33) Next, let's import the blank spaces we just generated. Click on *File > Import > CSV / Tab delimited*.



- (34) Select the TSV file you downloaded (`kia-orana-rehearsal.tsv`) and click on OK. Set the first column to *Begin time*, the second column to *End time* and the third column to "Annotation". Specify that the "first row of data" is number 2. Then click on "OK".



- (35) You now have a computer-generated transcription of the audio. We didn't train for very long, so the transcription itself is not very good, but it is a demonstration of what you can do with a speech recognition model.



- (36) Let's do one final exercise. Go back to Google Drive and open the file `inference-transcribe-w2v2-from-user.ipynb`. This file uses the model to transcribe a recording from the user (in this case, you!) Run the first box of code to connect to Google Drive. Run the second box of code to select a checkpoint for the transcription, and then click on "Confirm selection".

Pick your folder and checkpoint, then Confirm.

Sandboxes folder: `202606-cim-asr`

Model checkpoint: `sandbox-user / checkpoint-checkpoint-1224`

1 checkpoint(s) across: `sandbox-user`

→ will use checkpoint `checkpoint-checkpoint-1224` from sandbox `sandbox-user`

[Confirm selection](#)

Run the rest of the code. You will eventually arrive at these two boxes. They will help you record your voice. Run them.

```
# GET READY!
# When you run the next box, the computer will say that it's listening to you.
# I recommend that you say something simple. For example:
#
# Kia orana kotou katoatoa!
#
# It means "Hello" or "How are you?"
# After you said it, click on "Press to stop". Once the "play" button (on the left) stops rotating, that's
# when your recording will be ready and you can continue to the next step. (The button will keep saying
# "saving the recording... pls wait!". Don't pay attention to that. Pay attention to the play button. If
# the rotating line has stopped, then you're good).
#
# The system will make many mistakes because it's only trained on 16 minutes of audio.
# As the system learns from more data, the transcriptions will get better.

#-----
# This will open a button to record audio.
# It will save the file as temp.wav
# Once you see a play button, this means that the file is saved correctly.
#-----

%rm /content/temp.wav
%cd /content/
x, sr = get_audio()
```

The second box will ask you for permission to use the microphone. Grant it permission.

This cell is requesting browser permissions

You can accept the following browser permissions in this session:

- Microphone

After you accept the browser permissions, re-run the cell to have the permissions take effect.

[Cancel](#) [Accept](#)

Allow colab.research.google.com to use your microphone?

Microphone Array (Intel® Smart Sound Technology for Digital

☐ Remember for all microphones

[Allow](#) [Block](#)

- (37) Immediately after you've granted the permission, record yourself saying something in Cook Islands Maori. I recommend saying:

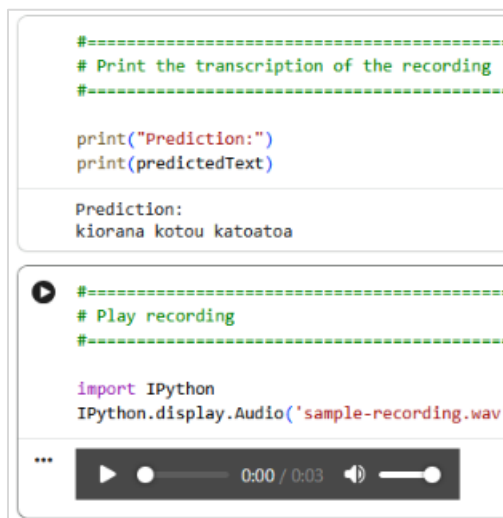
Kia orana kotou katoatoa!

This phrase means "Hi, everyone!". Then, click on "Recording... press to stop" to finish recording.



```
#-----  
# This will open a button to record audio.  
# It will save the file as temp.wav  
# Once you see a play button, this means that the file is saved correctly.  
#-----  
  
%rm /content/temp.wav  
%cd /content/  
x, sr = get_audio()  
  
*** rm: cannot remove '/content/temp.wav': No such file or directory  
/content  
Recording... press to stop
```

Run the rest of the code. This will produce a transcription of the recording.



```
#-----  
# Print the transcription of the recording  
#-----  
  
print("Prediction:")  
print(predictedText)  
  
Prediction:  
kiorana kotou katoatoa  
  
#-----  
# Play recording  
#-----  
  
import IPython  
IPython.display.Audio('sample-recording.wav')  
  
***
```

Congratulations! This is the end of the exercise.